

Python Code

```
1  import pandas as pd
2  import numpy as np
3  data = {
4      'Product_ID': [201, 202, 203, 204, 205],
5      'Product_Name': ['Laptop', 'Smartphone', 'Wireless Mouse', 'Smartwatch',
6      'Headphones'],
7      'Stock_Count': [15, 25, 50, 12, 40],
8      'Category': ['Electronics', 'Mobile', 'Accessories', 'Wearables', 'Accessories'],
9      'Price_INR': [55000, 28000, 1200, 4500, 2500]
10 }
11 df = pd.DataFrame(data)
12 print("Original DataFrame")
13 print(df)
14
15 print("\nShape of the given table: ", df.shape)
16
17 print("\nPrint first 3 rows using head")
18 print(df.head(3))
19
20 print("\nPrint last 2 rows using tail")
21 print(df.tail(2))
22
23 print("\nStatistical summary")
24 print(df.describe())
25
26 print("\nPrint particular col in reverse (Product Names)")
27 print(df['Product_Name'][::-1])
28
29 print("\nPrint particular multiple columns (First 2 rows)")
30 print(df[['Product_Name', 'Category'][:2])
31
32 print("\nPrint particular rows (Slicing)")
33 print(df[:2])
34
35 print("\nPrint particular row col detail using iloc (Row 0, Col 4)")
36 print("Price of first product:", df.iloc[0, 4])
37
38 print("\nPrint full row detail using iloc (Row 2)")
39 print(df.iloc[2])
40
41 print("\nPrint particular row using loc (Custom Index)")
42 df_loc = pd.DataFrame(data, index=['P1', 'P2', 'P3', 'P4', 'P5'])
43 print(df_loc.loc['P1':'P3'])
44
45 print("\nFiltering: Particular column filter (Category == Accessories)")
46 print(df[df['Category'] == 'Accessories'])
47
48 print("\nFiltering: Filter by range (Price > 4000)")
49 print(df[df['Price_INR'] > 4000])
```

```
50
51 print("\nSorting: Stock (Small to Large) and Product Name (Large to Small)")
52 print(df.sort_values(['Stock_Count', 'Product_Name'], ascending=[True, False]))
53
54 print("\nPrice Statistics")
55 print(f"Average Price: {df['Price_INR'].mean()}")
56 print(f"Median Price: {df['Price_INR'].median()}")
57 print(f"Minimum Price: {df['Price_INR'].min()}")
58 print(f"Maximum Price: {df['Price_INR'].max()}")
59 print(f"Standard Deviation of Price: {df['Price_INR'].std()}")
60 print(f"Total Value of Stock Prices: {df['Price_INR'].sum()}")
61
62 print("Add Column: Delivery Charges")
63 df['Delivery_Charges'] = [150, 100, 50, 80, 50]
64 print(df.to_string())
65
66
67 print("Convert Price to USD")
68 df['Price_USD'] = df['Price_INR'] / 83
69 print(df.to_string())
70
71
72 print("Adding Custom Discount based on Category")
73 df['Discount_Amount'] = [5000, 2000, 150, 400, 200]
74 print(df.to_string())
75
```

===== RESTART: C:\Python314\VIJAY-GEN AI-VL-7.1.py =====

Original DataFrame

	Product_ID	Product_Name	Stock_Count	Category	Price_INR
0	201	Laptop	15	Electronics	55000
1	202	Smartphone	25	Mobile	28000
2	203	Wireless Mouse	50	Accessories	1200
3	204	Smartwatch	12	Wearables	4500
4	205	Headphones	40	Accessories	2500

Shape of the given table: (5, 5)

Print first 3 rows using head

	Product_ID	Product_Name	Stock_Count	Category	Price_INR
0	201	Laptop	15	Electronics	55000
1	202	Smartphone	25	Mobile	28000
2	203	Wireless Mouse	50	Accessories	1200

Print last 2 rows using tail

	Product_ID	Product_Name	Stock_Count	Category	Price_INR
3	204	Smartwatch	12	Wearables	4500
4	205	Headphones	40	Accessories	2500

Add Column: Delivery Charges

	Product_ID	Product_Name	Stock_Count	Category	Price_INR	Delivery_Charges
0	201	Laptop	15	Electronics	55000	150
1	202	Smartphone	25	Mobile	28000	100
2	203	Wireless Mouse	50	Accessories	1200	50
3	204	Smartwatch	12	Wearables	4500	80
4	205	Headphones	40	Accessories	2500	50

Convert Price to USD

	Product_ID	Product_Name	Stock_Count	Category	Price_INR	Delivery_Charges	Price_USD
0	201	Laptop	15	Electronics	55000	150	662.650602
1	202	Smartphone	25	Mobile	28000	100	337.349398
2	203	Wireless Mouse	50	Accessories	1200	50	14.457831
3	204	Smartwatch	12	Wearables	4500	80	54.216867
4	205	Headphones	40	Accessories	2500	50	30.120482

	Product_ID	Product_Name	Stock_Count	Category	Price_INR	Delivery_Charges	Price_USD	Discount
0	201	Laptop	15	Electronics	55000	150	662.650602	5000
1	202	Smartphone	25	Mobile	28000	100	337.349398	2000
2	203	Wireless Mouse	50	Accessories	1200	50	14.457831	150
3	204	Smartwatch	12	Wearables	4500	80	54.216867	400
4	205	Headphones	40	Accessories	2500	50	30.120482	200

Filtering: Particular column filter (Category == Accessories)

	Product_ID	Product_Name	Stock_Count	Category	Price_INR
2	203	Wireless Mouse	50	Accessories	1200
4	205	Headphones	40	Accessories	2500

Filtering: Filter by range (Price > 4000)

	Product_ID	Product_Name	Stock_Count	Category	Price_INR
0	201	Laptop	15	Electronics	55000
1	202	Smartphone	25	Mobile	28000
3	204	Smartwatch	12	Wearables	4500

Sorting: Stock (Small to Large) and Product Name (Large to Small)

	Product_ID	Product_Name	Stock_Count	Category	Price_INR
3	204	Smartwatch	12	Wearables	4500
0	201	Laptop	15	Electronics	55000
1	202	Smartphone	25	Mobile	28000
4	205	Headphones	40	Accessories	2500
2	203	Wireless Mouse	50	Accessories	1200

Add Column: Delivery Charges

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1	202	Smartphone	25	Mobile	28000	100	337.349398	2000
2	203	Wireless Mouse	50	Accessories	1200	50	14.457831	150
3	204	Smartwatch	12	Wearables	4500	80	54.216867	400
4	205	Headphones	40	Accessories	2500	50	30.120482	200